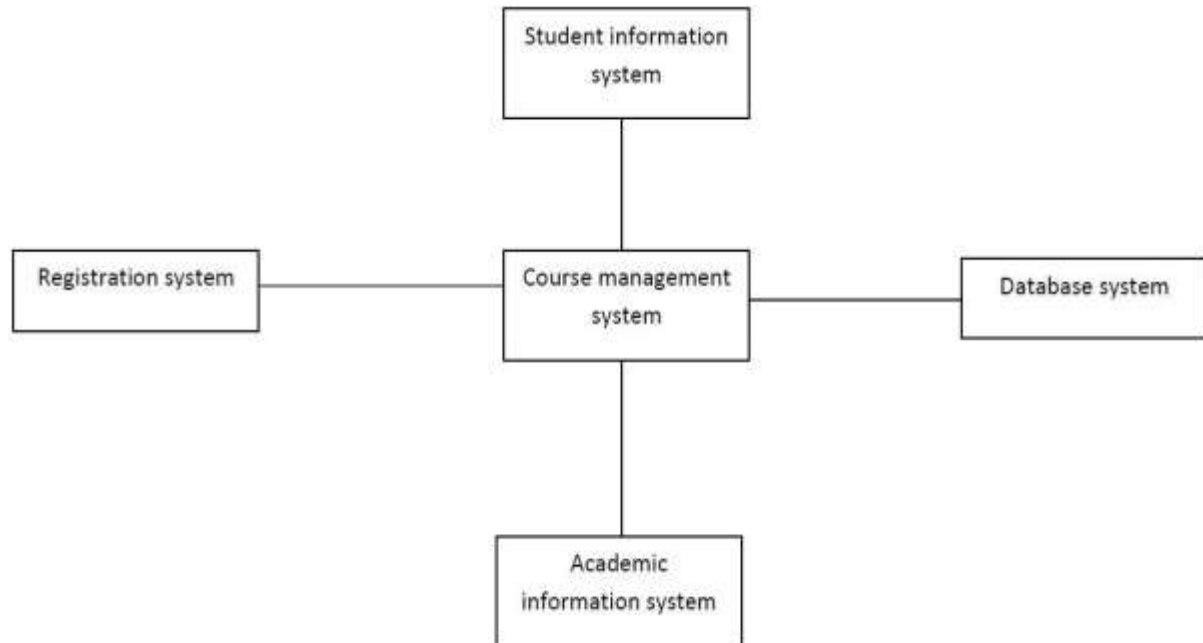


DATA MODELING and DESIGN:

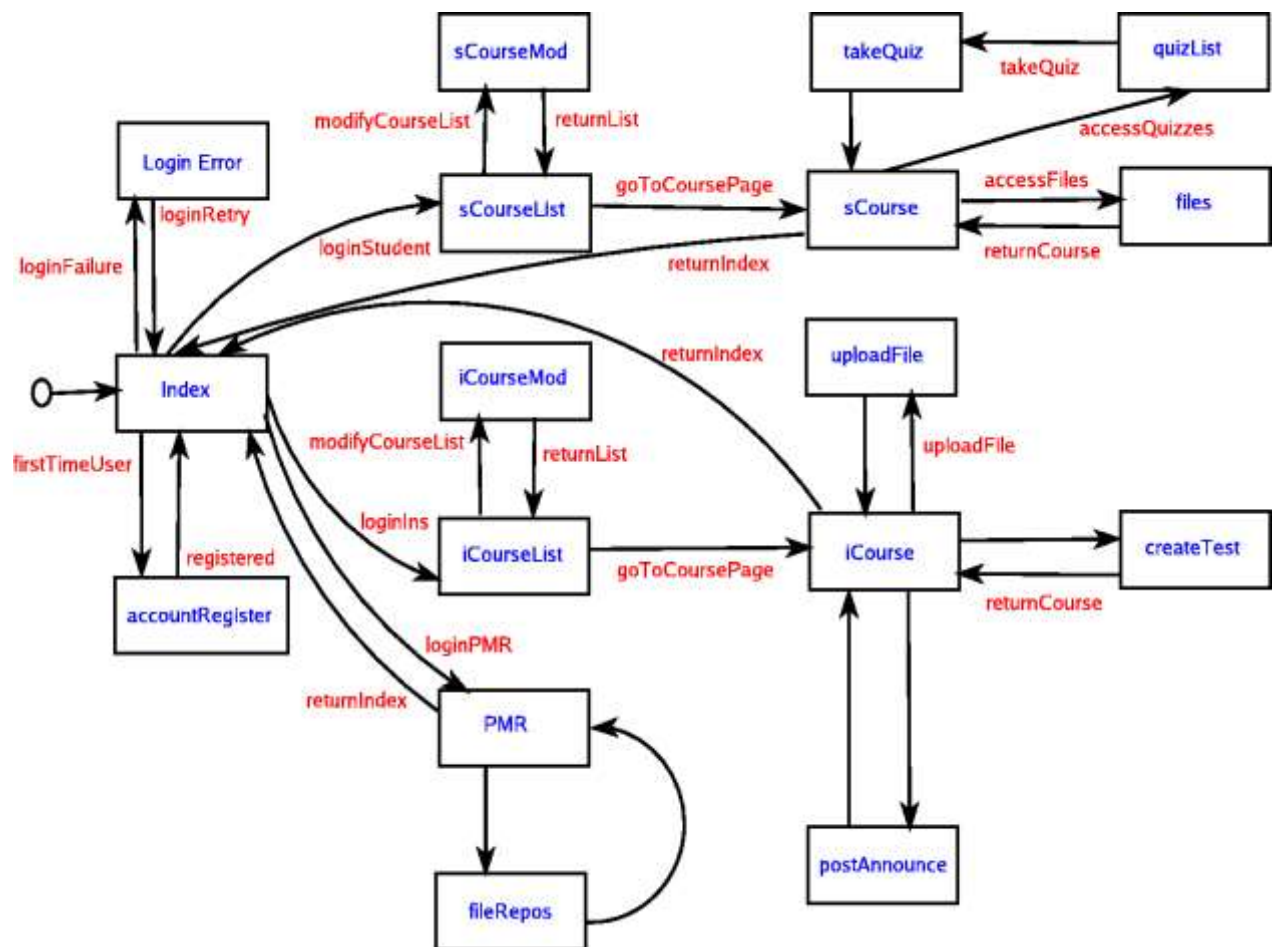
(1)Product Perspective

The system will be operating within university environment. This environment has another system that will interact with this system so we need interfaces between these systems



(2) Flow Chart

The below diagram will provide the overall flow of the project.



(3) Data Dictionary

(3.1) StudentDetails

FIELD NAME	TYPE	CONSTRAINTS
Sid	Varchar2	Primary key
Name	Varchar2	
Roll_No	Varchar2	Notnull
Regulation	Varchar	
Courseid	Number	Foreign key
grade	Char	
Fid	Varchar2	Foreign Key

(3.2)CourseDetails

FIELD NAME	TYPE	CONSTRAINTS
Courseid	Number	Primary key
CourseName	Varchar2	
Start_date	Date	
End_date	Date	
Subject	Varchar2	not null

(3.3)FacultyDetails

FIELD NAME	TYPE	CONSTRAINTS
Fid	Varchar2	Primary key
Name	Varchar2	
Courseid	Number	Foreign Key
Designation	Varchar	
Subject	Varchar	

(3.4)LoginDetails

FIELD NAME	TYPE	CONSTRAINTS
Userid	Varchar2	Unique
Password	Varchar2	Not null

Software Designing

UML

UML stands for Unified Modeling Language. This object-oriented system of notation has evolved from the work of Grady Booch, James Rum Baugh, Ivar Jacobson, and the Rational Software Corporation. These renowned computer scientists fused their respective technologies into a single, standardized model. Today, UML is accepted by the Object Management Group (OMG) as the standard for modeling object oriented programs.

UML Diagrams:

UML defines nine types of diagrams: class (package), object, use case, sequence, collaboration, state chart, activity, component, and deployment diagram.

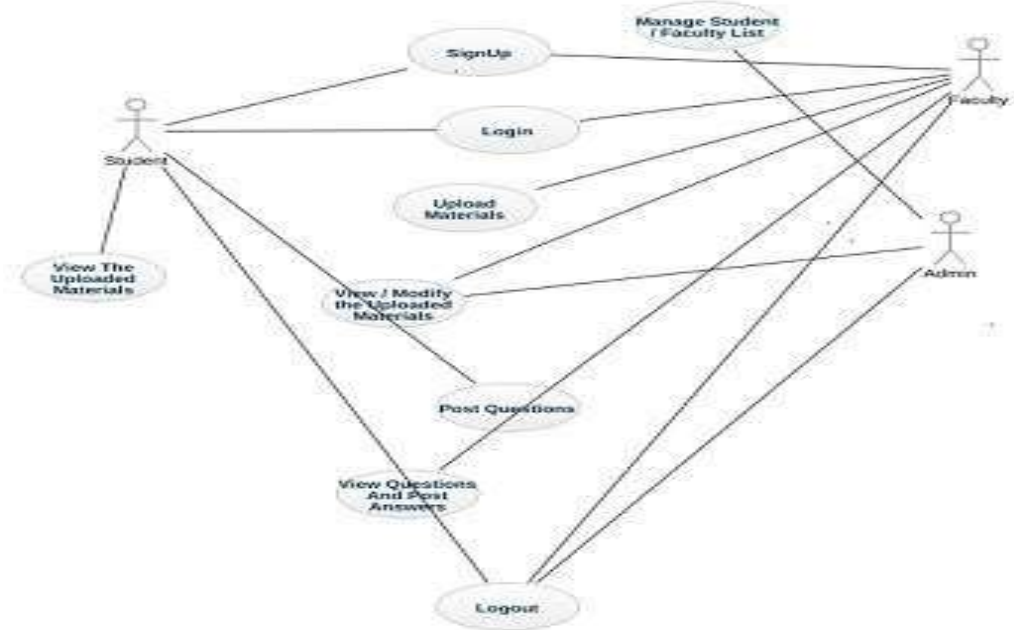
(1)Use Case Diagram

Use case diagrams are used to gather the requirements of a system including internal and external influences. These requirements are mostly design requirements. Hence, when a

system is analyzed to gather its functionalities, use cases are prepared and actors are identified.

The purposes of use case diagrams can be defined as follows –

- Used to gather the requirements of a system.
- Used to get an outside view of a system.
- Identify the external and internal factors influencing the system.
- Show the interaction among the requirements is actors.

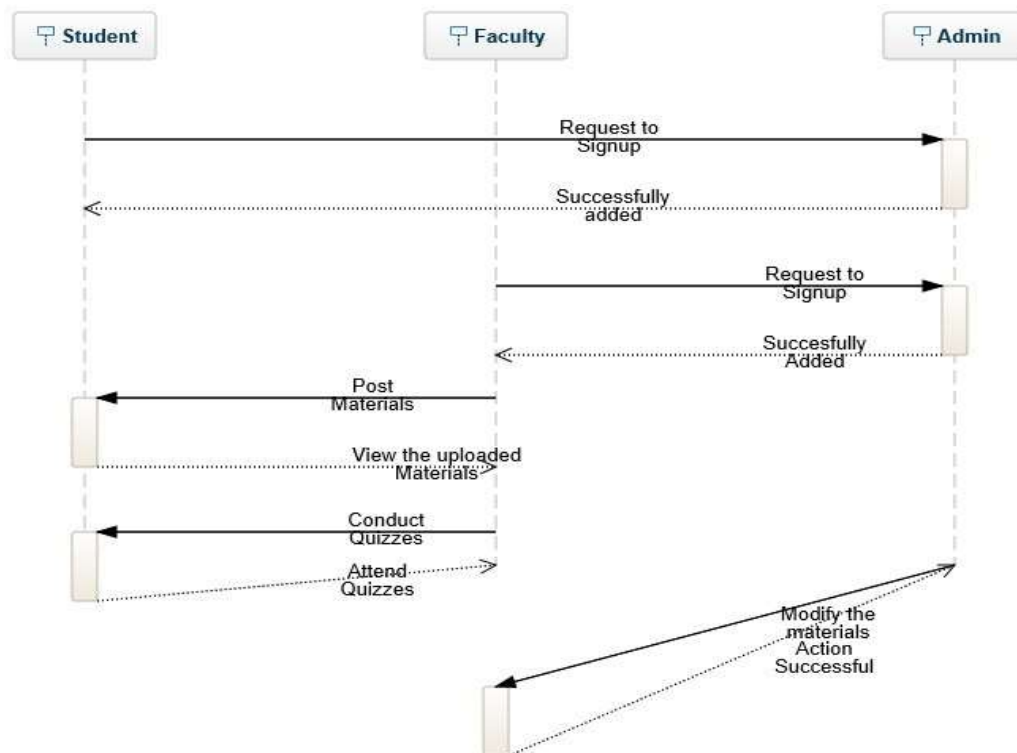


Sequence Diagram

This interactive behavior is represented in UML by Sequence **diagram**. Sequencediagram emphasizes on time sequence of messages that send and receive messages.

Following things are to be identified clearly before drawing the sequence diagram

- Objects taking part in the interaction.
- Message flows among the objects.
- The sequence in which the messages are flowing.
- Object organization.

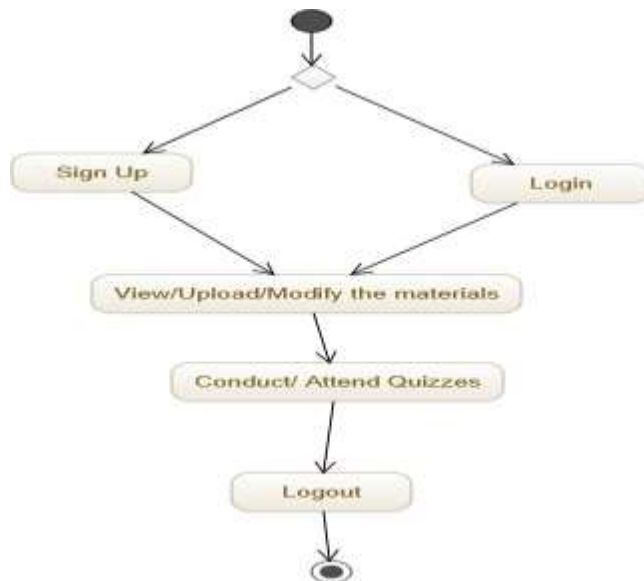


Activity Diagram

The basic purposes of activity diagrams are to capture the dynamic behavior of the system. Activity is a particular operation of the system. Activity diagrams are not only used for visualizing the dynamic nature of a system, but they are also used to construct the executable system by using forward and reverse engineering techniques. The only missing thing in the activity diagram is the message part.

The purpose of an activity diagram can be described as –

- Draw the activity flow of a system.
- Describe the sequence from one activity to another.
- Describe the parallel, branched and concurrent flow of the system.



Class Diagram

The purpose of class diagram is to model the static view of an application. Class diagrams are the only diagrams which can be directly mapped with object-oriented languages and thus widely used at the time of construction.

The purpose of the class diagram can be summarized as –

- Analysis and design of the static view of an application.
- Describe responsibilities of a system.
- Base for component and deployment diagrams.
- Forward and reverse engineering.

